

HW3

#1.

$$\textcircled{a} X_1(z) = \sum_{n=-\infty}^{\infty} x_1[n] z^{-n} = \sum_{n=0}^{\infty} (0.4 z^{-1})^n = \frac{1}{1-0.4 z^{-1}}, \quad |z| > 0.4$$

$$\textcircled{b} X_2(z) = \sum_{n=0}^{\infty} (-0.6 z^{-1})^n = \frac{1}{1+0.6 z^{-1}}, \quad |z| > 0.6$$

$$\textcircled{c} X_3(z) = \sum_{n=4}^{\infty} (0.3 z^{-1})^n = \frac{0.3^4}{1-0.3 z^{-1}} + \sum_{n=0}^3 (0.3 z^{-1})^n, \quad |z| > 0.3$$

$$\textcircled{d} X_4(z) = \sum_{n=-\infty}^{-2} (-0.3 z^{-1})^n = \frac{0.3^2}{1+0.3 z^{-1}} + \sum_{n=-1}^0 (-0.3 z^{-1})^n, \quad |z| < 0.3$$

$$\textcircled{e} R_{y1} = R_1 \cap R_2 = |z| > 0.6$$

$$\textcircled{f} R_{y2} = R_3 \cap R_4 = \emptyset$$

#2.

$$\frac{z^2 - z}{(z+1)(z+\frac{1}{3})} = A + \frac{B}{z+1} + \frac{C}{z+\frac{1}{3}}$$

$A=1$
$B=-3$
$C=-2$

$$Y(z) = \frac{1-z^{-1}}{(1+z^{-1})(1+\frac{1}{3}z^{-1})} \stackrel{A}{=} \frac{A}{1+z^{-1}} + \frac{B}{1+\frac{1}{3}z^{-1}}$$

Where $A=3, B=-2$.

(i) $|z| > 1$:

$$y[n] = (-3)(-1)^n u[n] - 2(-\frac{1}{3})^n u[n]$$

(ii) $|z| < \frac{1}{3}$:

$$y[n] = -3(-1)^n u[-n-1] + 2(-\frac{1}{3})^n u[-n-1]$$

(iii) $\frac{1}{3} < |z| < 1$:

$$y[n] = -3(-1)^n u[-n-1] - 2(-\frac{1}{3})^n u[n]$$

#3.

① Consider $H(e^{j\omega}) = \frac{1}{1 - 2r\cos\theta e^{-j\omega} + r^2 e^{-j2\omega}}$, ~~then~~ $H(e^{j\omega}) = H(z)|_{z=e^{j\omega}}$, is the original formula. So $H(z)$.

$$H(e^{j\omega}) = \frac{1}{[1 - (re^{j\theta})e^{-j\omega}][1 - (re^{-j\theta})e^{-j\omega}]}$$

$$= \frac{A}{1 - (re^{j\theta})e^{-j\omega}} + \frac{B}{1 - (re^{-j\theta})e^{-j\omega}}$$

$$\text{Where } A = \frac{e^{j\theta}}{2j\sin\theta}, \quad B = -\frac{e^{-j\theta}}{2j\sin\theta}$$

$$\Rightarrow h[n] = [A \cdot (re^{j\theta})^n + B (re^{-j\theta})^n] u[n]$$

$$= r^n \left[\frac{e^{j\theta(n+1)}}{2j\sin\theta} - \frac{e^{-j\theta(n+1)}}{2j\sin\theta} \right] u[n]$$

$$= r^n \cdot \frac{\sin[(n+1)\theta]}{\sin\theta} \cdot u[n]$$

(b).

$$H(z) = \frac{z-2}{(z+0.4)(z-0.2)} = \frac{4}{z+0.4} + \frac{-3}{z-0.2} = \frac{4}{0.4} \left(1 - \frac{1}{1+0.4z^{-1}}\right) + \frac{-3}{-0.2} \left(1 - \frac{1}{1-0.2z^{-1}}\right)$$

$$= 25 + \frac{-10}{1+0.4z^{-1}} + \frac{-15}{1-0.2z^{-1}}$$

(i) $|z| > 0.4$

$$\circ \quad x[n] = 25\delta[n] - 10(-0.4)^n u[n] - 15(0.2)^n u[n]$$

(ii) $|z| < 0.2$

$$H(z) = \frac{10}{1+2.5z} + \frac{15}{1-5z} = 10 \sum_{n=0}^{\infty} (-2.5z)^{-n} + 15 \sum_{n=0}^{\infty} (5z)^{-(-n)}$$

$$= 10 \sum_{n=-\infty}^0 (-0.4z^{-1})^n + 15 \sum_{n=-\infty}^0 (0.2z^{-1})^n$$

$$\Rightarrow x[n] = 10(-0.4)^n u[-n] + 15(0.2)^n u[-n] = 25\delta[n] + 10(-0.4)^n u[-n-1] + 15(0.2)^n u[-n-1]$$

(iii) $0.2 < |z| < 0.4$

$$\circ \quad H(z) = 15 + \frac{-15}{1-0.2z^{-1}} + 10 \sum_{n=-\infty}^0 (-0.4z^{-1})^n$$

$$x[n] = 15\delta[n] - 15(0.2)^n u[n] + 10(-0.4)^n u[-n]$$

$$= 25\delta[n] - 15(0.2)^n u[n] + 10(-0.4)^n u[-n-1]$$

#4 $\tau(\omega) = -\frac{d\phi(\omega)}{d\omega} = -\frac{d}{d\omega} \{ \angle H(e^{j\omega}) \} \dots (1)$

$$\operatorname{Re} \left\{ \frac{j \frac{dH(e^{j\omega})}{d\omega}}{H(e^{j\omega})} \right\} = -\operatorname{Im} \left\{ \frac{\frac{dH}{d\omega}}{H} \right\} \dots (2)$$

$$H \triangleq A + jB, \quad A, B \in \mathbb{R}$$

$$-\operatorname{Im} \left\{ \frac{\frac{dH}{d\omega}}{H} \right\} = -\operatorname{Im} \left\{ \frac{\frac{dA}{d\omega} + j \frac{dB}{d\omega}}{A + jB} \right\} = -\operatorname{Im} \left\{ \frac{(\frac{dA}{d\omega} + j \frac{dB}{d\omega})(A - jB)}{A^2 + B^2} \right\}$$

$$= \frac{-1}{|H|^2} \left(-B \frac{dA}{d\omega} + A \frac{dB}{d\omega} \right) = \frac{1}{|H|^2} \left(B \frac{dA}{d\omega} - A \frac{dB}{d\omega} \right)$$

$$\angle H = \tan^{-1} \left(\frac{B}{A} \right) = \frac{1}{1 + \left(\frac{B}{A} \right)^2} \cdot \frac{dB}{A} \quad \text{(Note: } dA \text{ is circled in the original image)}$$

$$-\frac{d\angle H}{d\omega} = \frac{-1}{1 + \left(\frac{B}{A} \right)^2} \cdot \frac{A \frac{dB}{d\omega} - B \frac{dA}{d\omega}}{A^2} = \frac{1}{A^2 + B^2} \left(B \frac{dA}{d\omega} - A \frac{dB}{d\omega} \right)$$

$$\therefore (1) = (2)$$

QED.

#5.

(a) 1^o Symmetrical

$$a_1 = a_7, a_2 = a_6, a_3 = a_5$$

2^o ~~Asymmetrical~~ Anti-symmetrical:

$$a_1 = -a_7, a_2 = -a_6, a_3 = -a_5, a_4 = 0.$$

(b) $y[n]$ should be even

$\therefore k=2$. (Properties of DFT) Then, $H(e^{j\omega}) = a_3 + 2a_2 \cos \omega + 2a_1 \cos 2\omega$.

Can be

#6

$$H_1(e^{j\omega}) = \alpha + e^{-j\omega}$$

$$H_2(e^{j\omega}) = \frac{1}{1 - \beta e^{-j\omega}}$$

$$H(e^{j\omega}) = H_1 H_2 = \frac{\alpha + e^{-j\omega}}{1 - \beta e^{-j\omega}}$$

$$|H|^2 = \frac{\alpha^2 + 2\alpha \cos \omega + 1}{(1 - \beta e^{-j\omega})(1 - \beta e^{j\omega})} = \frac{\alpha^2 + 2\alpha \cos \omega + 1}{1 - 2\beta \cos \omega + \beta^2} = 1$$

$$\Rightarrow (2\alpha + 2\beta) \cos \omega = \beta^2 - \alpha^2$$

$$\Rightarrow \alpha + \beta = 0$$

#7. $H(e^{j\omega}) = e^{-j\omega \frac{N-1}{2}} e^{j\frac{\pi}{2}} \sum_{n=1}^{(N-1)/2} c[n] \sin(\omega n)$, $c[n] = 2h[\frac{N-1}{2} - n] = 2h[2-n]$

$$\frac{1}{2} = |H(e^{j\frac{\pi}{4}})| = \sum_{n=1}^{N/2} c[n] \sin(\frac{\pi}{4} n) = 2h[1] \cdot \frac{1}{\sqrt{2}} + 2h[0] \cdot 1 = \sqrt{2} h[1] + 2h[0]$$

$$1 = |H(e^{j\frac{\pi}{2}})| = \sum_{n=1}^2 c[n] \sin(\frac{\pi}{2} n) = 2h[1] \cdot 1 \Rightarrow h[1] = \frac{1}{2}$$

$$\Rightarrow h[0] = \frac{1-\sqrt{2}}{4}$$

$$H(e^{j\omega}) = e^{-j\omega} e^{j\frac{\pi}{2}} \sum_{n=1}^2 2h[2-n] \sin(\omega n)$$

$$H(e^{j\frac{\pi}{4}}) = e^{-j\frac{\pi}{2}} e^{j\frac{\pi}{2}} \sum_{n=1}^2 2h[2-n] \sin(\frac{\pi}{4} n) = \sum_{n=1}^2 2h[2-n] \sin(\frac{\pi}{4} n)$$

$$\Rightarrow \frac{1}{2} = |2h[1] + 2h[0]|$$

$$|H(e^{j\frac{\pi}{2}})| = e^{-j\frac{\pi}{2}} \sum_{n=1}^2 2h[2-n] \sin(\frac{\pi}{2} n) \Rightarrow h[1] = \frac{1}{2}, h[0] = \frac{1-\sqrt{2}}{4}$$

$$\Rightarrow |H| = |2h[1]|$$

$$c[0] = 0 \Rightarrow h[2] = 0 \quad \therefore h[n] = \left\{ \frac{1-\sqrt{2}}{4}, \frac{1}{2}, 0, -\frac{1}{2}, \frac{\sqrt{2}-1}{4} \right\} \quad (0 \leq n \leq 4)$$

$$H(e^{j\omega}) = \frac{1-\sqrt{2}}{4} + \frac{1}{2}e^{-j\omega} - \frac{1}{2}e^{-j\omega 3} + \frac{\sqrt{2}-1}{4}e^{-j\omega 4}$$

$$2^o \quad h[0] = \frac{-1-\sqrt{2}}{4} \quad h[1] = \frac{1}{2}$$

$$H(e^{j\omega}) = \frac{-1-\sqrt{2}}{4} + \frac{1}{2}e^{-j\omega} - \frac{1}{2}e^{-j\omega 3} + \frac{1+\sqrt{2}}{4}e^{-j\omega 4}$$

$$3^o \quad h[0] = \frac{\sqrt{2}-1}{4} \quad h[1] = -\frac{1}{2}$$

$$H(e^{j\omega}) = \frac{\sqrt{2}-1}{4} - \frac{1}{2}e^{-j\omega} + \frac{1}{2}e^{-j\omega 3} + \frac{1-\sqrt{2}}{4}e^{-j\omega 4}$$

$$4^o \quad h[0] = \frac{\sqrt{2}+1}{4} \quad h[1] = -\frac{1}{2}$$

$$H(e^{j\omega}) = \frac{\sqrt{2}+1}{4} - \frac{1}{2}e^{-j\omega} + \frac{1}{2}e^{-j\omega 3} + \frac{\sqrt{2}-1}{4}e^{-j\omega 4}$$

#8. $\frac{x[n]}{h[n]}$ real $\Leftrightarrow \Rightarrow X(e^{j\omega}) = X^*(e^{-j\omega})$

$$H(e^{j\omega}) = a_0 + a_1 e^{-j\omega} + a_2 e^{-j\omega 2} + a_3 e^{-j\omega 3}$$

$$\begin{cases} 0 = a_0 + a_1 + a_2 + a_3 \\ 7-j3 = a_0 + (-ja_1) + (-a_2) + ja_3 \\ 0 = a_0 - a_1 + a_2 - a_3 \end{cases}$$

$$H(e^{j\omega}) = H^*(e^{-j\omega})$$

$$\Rightarrow 7-j3 = H(e^{j\frac{\pi}{2}}) = H^*(e^{-j\frac{\pi}{2}}) = a_0 + ja_1 + (-a_2) - ja_3$$

$$\therefore \begin{bmatrix} 0 \\ 7-j3 \\ 0 \\ 7+j3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0.1207 \\ 0.0517 \\ -0.1207 \\ -0.0517 \end{bmatrix}$$

$$\therefore H(z) = 0.1207 + 0.0517z^{-1} - 0.1207z^{-2} - 0.0517z^{-3}$$

#9.

a.

$$H(z) = -2 + \frac{8}{1+0.5z^{-1}} + \frac{2}{1-0.4z^{-1}}$$

$$\Rightarrow h[n] = -2\delta[n] + 8(-0.5)^n u[n] + 2(0.4)^n u[n]$$

b.

$$X(z) = \frac{1.2}{1+0.2z^{-1}} - \frac{0.2}{1-0.3z^{-1}}$$

$$Y(z) = X(z)H(z) = \frac{(0.4z^2 - 2z + 8)(-2z^{-1} + 5)}{5(1+0.2z^{-1})(1-0.3z^{-1})(1+0.5z^{-1})(1-0.4z^{-1})}$$

$$= \frac{40}{1+0.2z^{-1}} + \frac{-92}{1+0.3z^{-1}} + \frac{60}{1+0.5z^{-1}}$$

$$\Rightarrow y[n] = 40(-0.2)^n u[n] - 92(-0.3)^n u[n] + 60(-0.5)^n u[n]$$

#10

$$Y - 0.8z^{-1}Y + 0.15z^{-2}Y = X$$

$$\Rightarrow H = \frac{Y}{X} = \frac{1}{1 - 0.8z^{-1} + 0.15z^{-2}}$$

(b)

$$H(z) = \frac{1}{(1 - 0.3z^{-1})(1 - 0.5z^{-1})} = \frac{-1.5}{1 - 0.3z^{-1}} + \frac{2.5}{1 - 0.5z^{-1}}$$

$$h[n] = -1.5(0.3)^n u[n] + 2.5(0.5)^n u[n]$$

(c)

$$u[n] \xleftrightarrow{Z} \frac{1}{1 - z^{-1}}$$

$$Y(z) = H(z) \cdot \frac{1}{1 - z^{-1}} = \frac{2.8571}{1 - z^{-1}} + \frac{-2.5}{1 - 0.5z^{-1}} + \frac{0.6429}{1 - 0.3z^{-1}}$$

$$\hat{y}[n] = 2.8571 u[n] - 2.5(0.5)^n u[n] + 0.6429(0.3)^n u[n]$$

#11

$$H(z) = (E_0 + z^{-1}E_1)(R_0 + z^{-1}R_1) + (E_0 - z^{-1}E_1)(-R_0 + z^{-1}R_1)$$

$$= z^{-1}(2E_0R_1 + 2E_1R_0) - z^{-2}(E_1R_1)$$

$y[n]$ is a scaled and delayed replica of $x[n]$, when $H(z) = Kz^{-n_0}$, K is a constant, $n_0 \in \mathbb{N}^*$

cb) $H_0 = 1 + \alpha z^{-1} \Rightarrow E_0 = 1, E_1 = \alpha$

$$H(z) = 2z^{-1}(R_1 + \alpha R_0) - \alpha z^{-2}R_1$$

1° delay 1:

$$\begin{cases} \alpha R_1 = 0 \\ R_1 + \alpha R_0 \neq 0 \end{cases} \Rightarrow \begin{cases} \text{(I) } \alpha = 0: R_1 = \text{constant} \neq 0 \quad \text{--- (1)} \\ \text{(II) } \alpha \neq 0: R_1 = 0 \\ R_0 = \text{constant} \neq 0 \quad \text{--- (2)} \end{cases}$$

2° delay 2:

$$\begin{cases} \alpha R_1 \neq \text{constant} \neq 0 \\ R_1 + \alpha R_0 = 0 \end{cases} \Rightarrow \begin{cases} \text{(I) } \alpha = 0: \text{ impossible} \\ \text{(II) } \alpha \neq 0: R_1 = -\alpha R_0 = \text{constant} \quad \text{--- (3)} \end{cases}$$

(C) 3 cases have been shown.